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अन्तरिक्ष प्रतिष्ठान नेपाल



Slippers2Sat-2 (S2S-2): Antenna Radiation Pattern

S2S-2 Team

Kathmandu, Nepal.



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Introduction

Uplink Frequency: 435.313 MHz

Downlink Frequency: 437.375 MHz

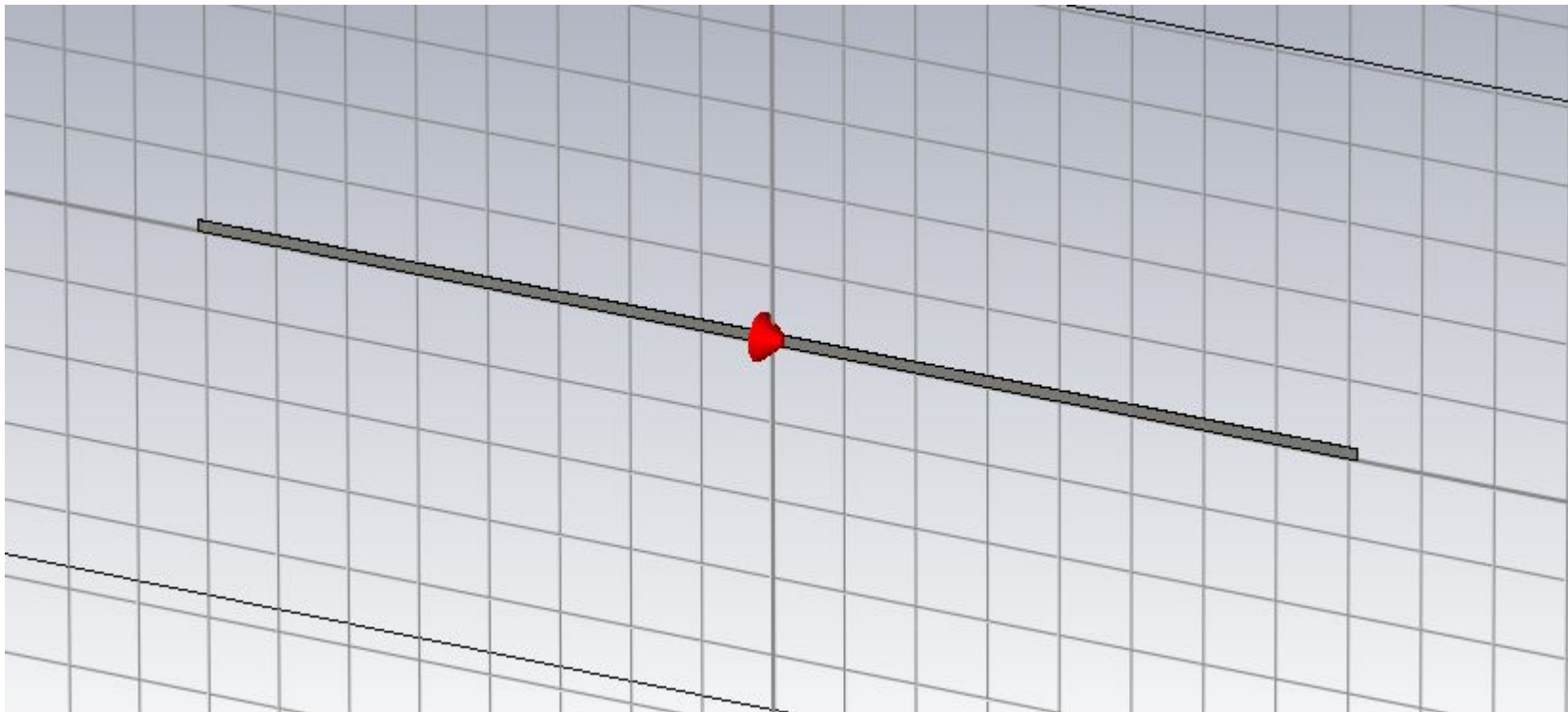




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Dipole Antenna

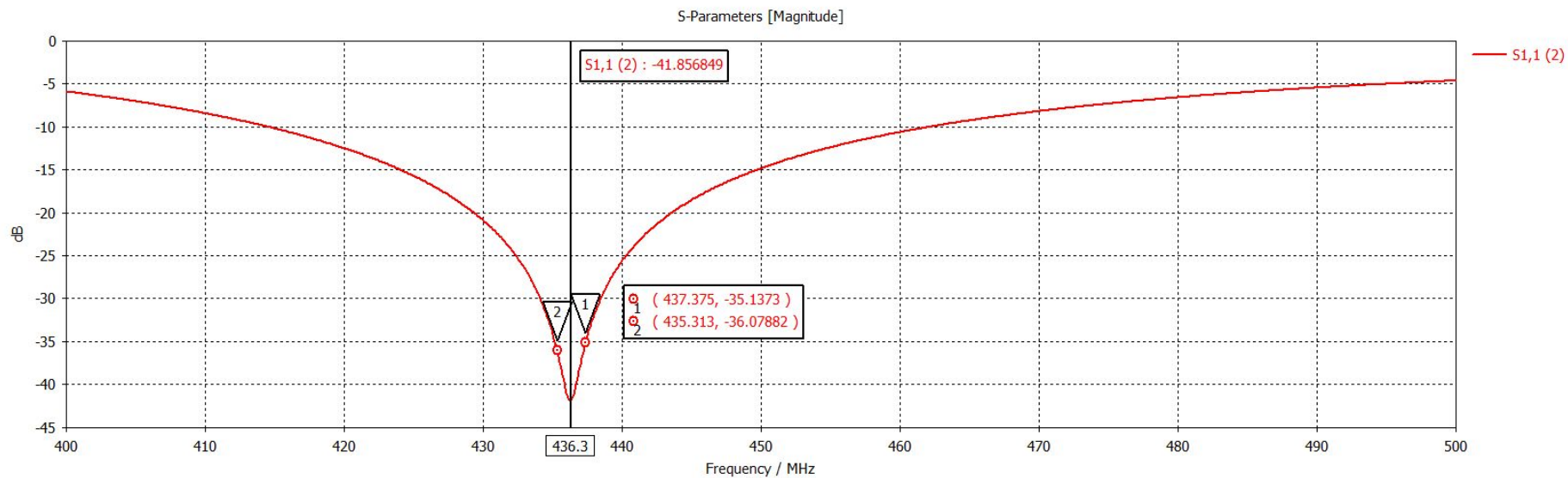




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S11 Parameter

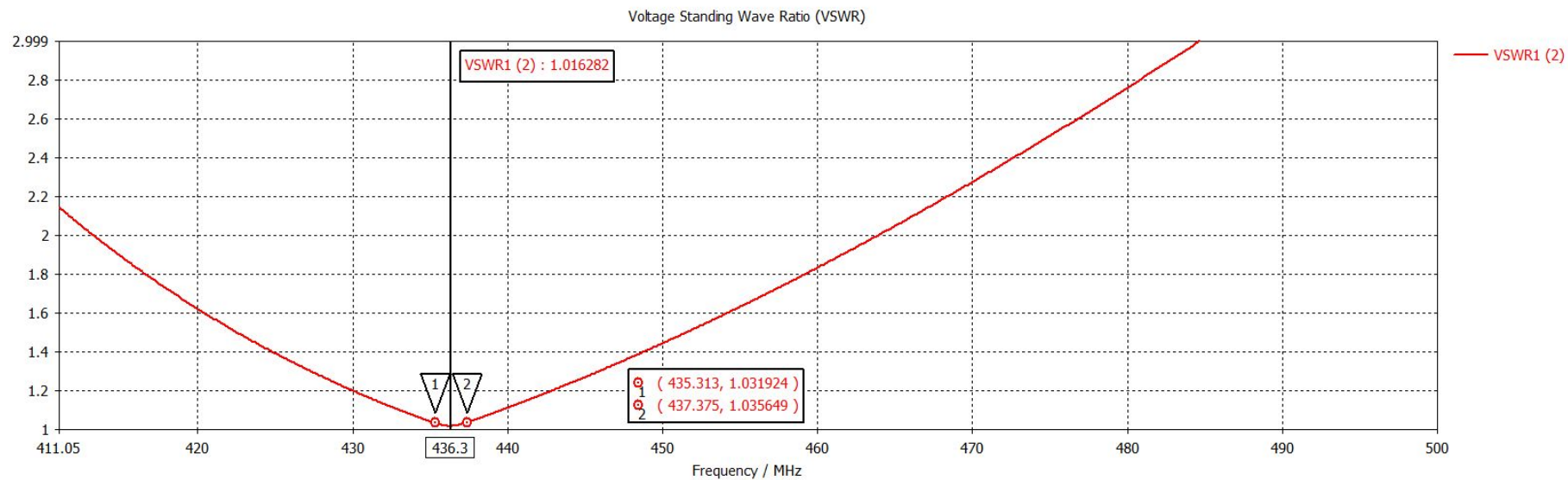




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VSWR

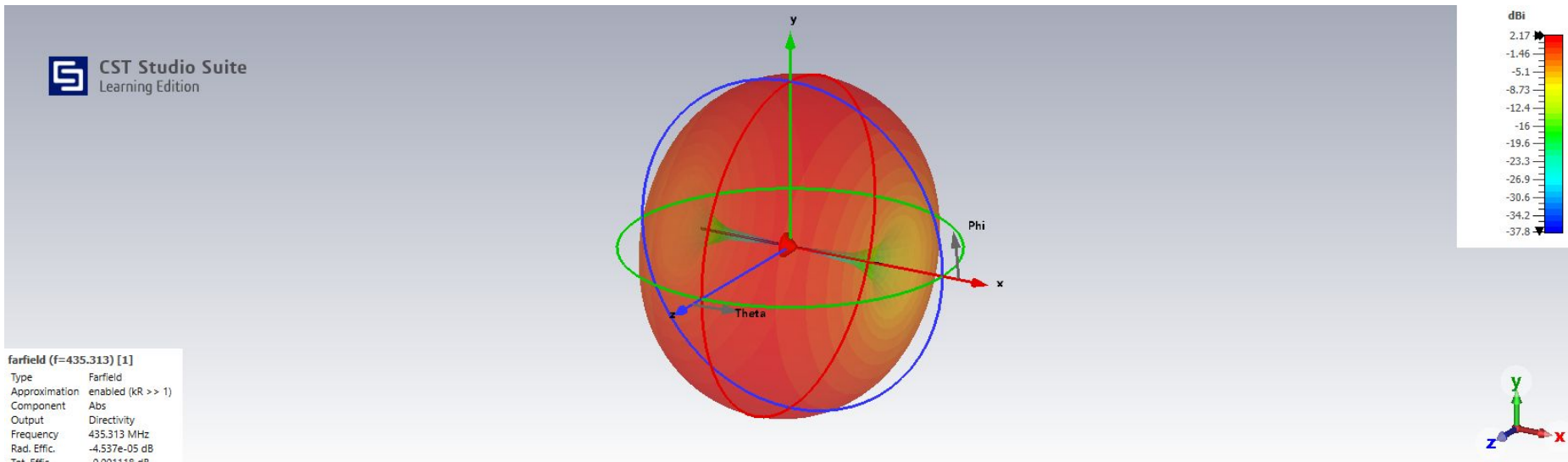




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3D-Radiation Pattern



Uplink Frequency: 435.313 MHz



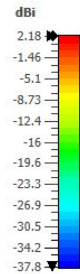
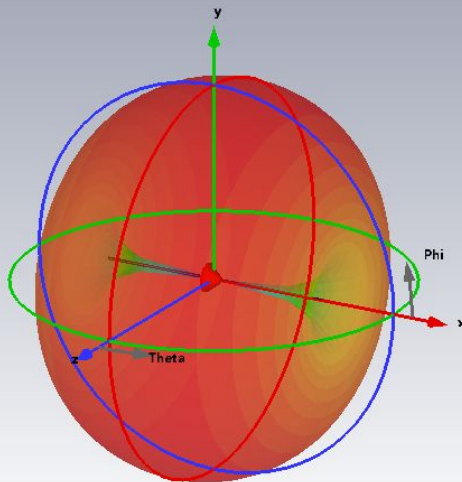
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3D-Radiation Pattern



CST Studio Suite
Learning Edition



farfield (f=437.375) [1]
Type Farfield
Approximation enabled (kR >> 1)
Component Abs
Output Directivity
Frequency 437.375 MHz
Rad. Effic. -3.759e-05 dB
Tot. Effic. -0.001371 dB
Dir. 2.177 dBi

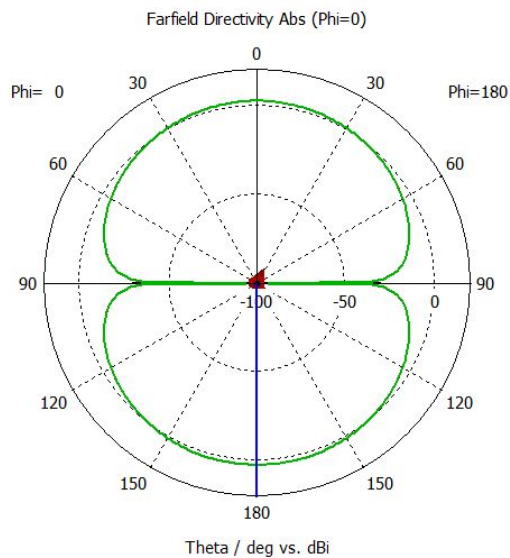
Downlink Frequency: 437.375 MHz



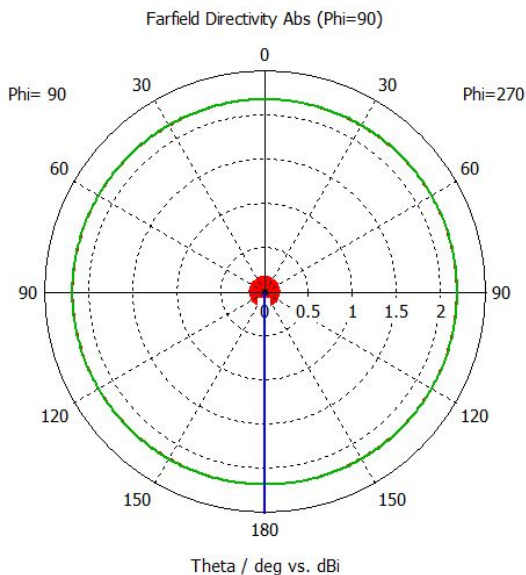
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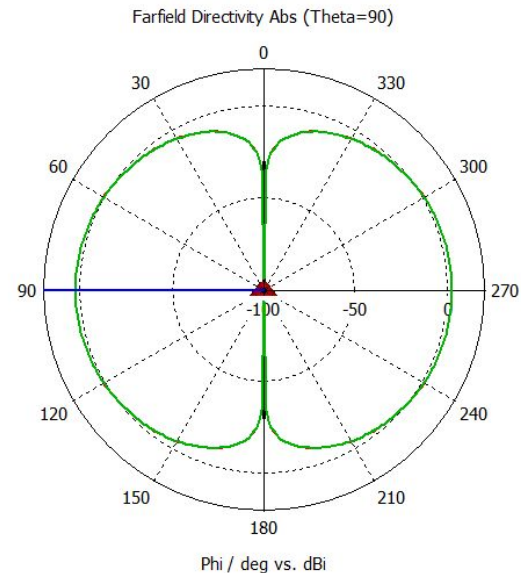
2D-Radiation Pattern



$\text{PHI} = 0$



$\text{PHI} = 90$



$\text{Theta} = 90$

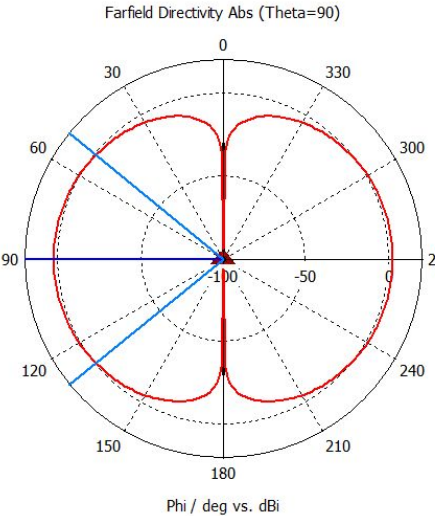
— farfield (f=435.313)
— farfield (f=437.375)



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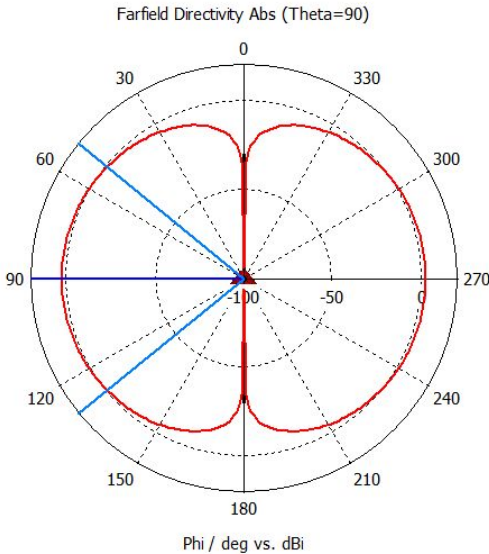
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Far-Field



— farfield (f=435.313) [1]

Frequency = 435.313 MHz
Main lobe magnitude = 2.17 dBi
Main lobe direction = 90.0 deg.
Angular width (3 dB) = 78.4 deg.



— farfield (f=437.375) [1]

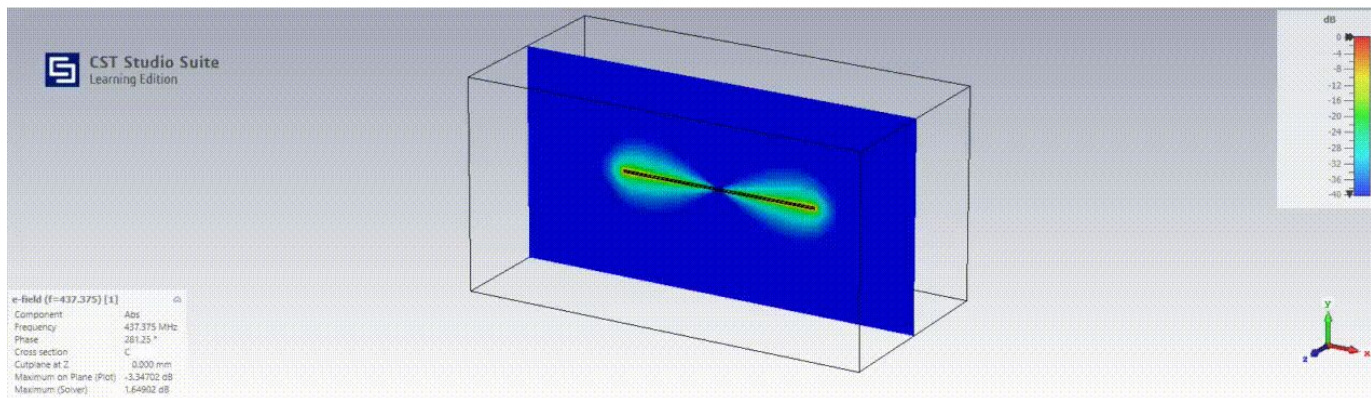
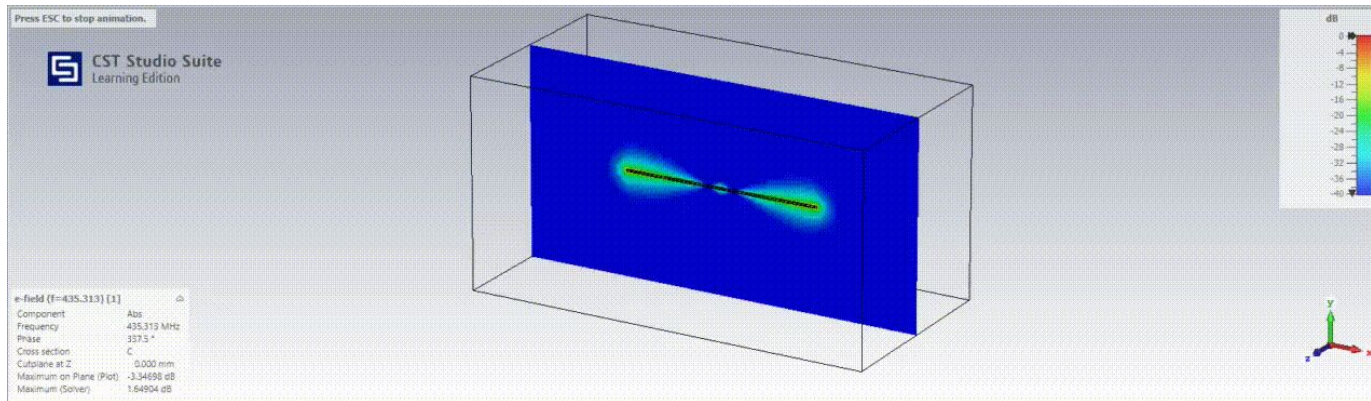
Frequency = 437.375 MHz
Main lobe magnitude = 2.18 dBi
Main lobe direction = 90.0 deg.
Angular width (3 dB) = 78.3 deg.



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E-Field Radiation





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H-Field Radiation

